

The Proto-Spinor: Conditional Spinorial Closure and the q79 Interface

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Version 8

Abstract

We isolate the part of the proto-spinor construction that is mathematically forced after its geometric premises are stated. A world-in-world field is a section of $\text{Hom}(TP, TI)$ for oriented rank-three bundles; its nine local components do not multiply manifold dimensions. If closure data must retain the nontrivial loop class in $SO(3)$, the faithful continuous double-cover carrier is spinorial and is represented through $\text{Spin}(3) \cong SU(2)$. This is a conditional lifting theorem, not a derivation of internal dimension three or of physical fermions from admissibility alone. For the selected q79 degree-three spectral cover we record the exact trace split of ranks $1 + 2$, the common $1 + 2 + 3$ carrier, and the local binary-dihedral lift of signed sheet monodromy. The shared circle is now a specified pullback of one universal flat $\mathbb{Z}_{64}$ differential line, not merely an isomorphic fiber. Its determinant, CLN, root-plane, and finite-Hessian actions intertwine exactly. The BHT and Hori fiber transforms share one orthogonal complex structure J_{FM} , and the binary sheet group has exactly two conjugate $\pm i$ phase-root lifts for which $(\pm i J_{\text{FM}})^2 = I$. Strict global Spin closure remains equivalent to a branch-relator obstruction calculation; the closed result is SpinC and finite symbolic. Circle, lens, and nil remain typed phase, finite-transport, and anchoring roles. The later trace-plane construction supplies a Clifford module, and the selected framed product source admits two equivalent binary-root factorizations of its complete free Dirac carrier and observable net. We explain the exact finite $864 \rightarrow 96$ reduction, conditional continuum and Deligne constructions, and selected finite family-response algebra without identifying dimensionless spectra with masses. A nonflat same-endpoint HYM comparison, interacting physical action, and strict source-selected magnitudes remain distinct downstream problems.

Revision note for version 8

Supersedes. Version 7; its revision notes remain below as an account of the earlier editions.

Reason. The finite-superconnection and SpinC sources continued beyond the older local-lift boundary. A result list did not explain their successive refinements or their different physical assumptions.

Resolution. Version 8 integrates finite branching and response operators, overlap and endpoint constructions, Deligne normalization, family-response results, and complete free-carrier root equivalence. The exploratory simulation is distinguished from a physical calculation. The curved connection compiler and nonlinear support-stratified strain shadow replace the excluded linear root-space route, with explicit scalar-field and polar-metric qualifications.

Retained. The conditional loop-memory theorem, shared line, finite return, and established finite/profile SM results are unchanged. No result is promoted by a packet’s success flag.

Open boundary. Strict global q79 Spin, nonflat HYM spectral and Hessian comparison, the selected interacting action and physical source values remain open. The framed free carrier, hidden projective HYM existence, and canonical operational quantum mechanics are not reopened.

Revision note for version 7

Supersedes. *The Proto-Spinor: Conditional Spinorial Closure and the q79 Interface*, version 6.

Reason. Version 6 contains the current finite SpinC and shared-line results, but the relation between loop memory, the double cover, the q79 finite return, and a physical fermion remained too easy to compress into one claim.

Resolution. Version 7 adds a four-layer reading guide, the explicit $2\pi/4\pi$ loop-memory picture, and a one-line derivation of the two conjugate finite returns. It keeps strict global Spin, continuum HYM intertwining, Lorentzian dynamics, and particle interpretation as separate gates and changes no theorem tier.

Retained result. The conditional double-cover theorem, universal flat shared line, finite Hessian naturality, and binary SpinC–Fourier return are retained unchanged.

Remaining boundary. Strict global q79 Spin, the nonflat FM/HYM lift, the strain-to-q79 continuum intertwiner, and a physical Lorentzian particle/action completion remain open.

Revision note for version 6

Supersedes. *The Proto-Spinor: Conditional Spinorial Closure and the q79 Interface*, version 5.

Reason. Version 5 stopped at a local Dic_3 lift and an untyped shared line. It did not include the later universal differential-line pullback, the common BHT–Hori Clifford operator, or the exact binary SpinC return.

Resolution. Version 6 retains the conditional double-cover theorem and adds the flat q79 shared-line theorem, finite Hessian naturality, the exact $J_{\text{FM}}^2 = -I$ polarization, and the two conjugate $\pm i$ phase-root lifts.

Retained result. Spinorial lifting is necessary when an oriented rank-three carrier must retain the nontrivial $SO(3)$ loop class.

Remaining boundary. Strict global q79 Spin, the physical nonflat FM/HYM lift, the strain-to-q79 continuum intertwiner, and a Lorentzian particle/action completion remain open gates.

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1 How to Read the Proto-Spinor

The construction has four layers, and the word “proto” is meant to keep them apart.

1. A local comparison field is a linear map between two oriented rank-three bundles. Its 3×3 matrix contains orientation and strain components.
2. Requiring the state carrier to remember the nontrivial loop in the orientation group replaces $SO(3)$ by its double cover $\text{Spin}(3) \cong SU(2)$.
3. The selected q79 finite geometry supplies a separate rank-1 + 2 + 3 carrier, a shared flat differential line, signed sheet monodromy, and an exact finite SpinC–Fourier return.
4. A physical fermion requires a Lorentzian spinor bundle, Dirac dynamics, statistics, interactions, observables, and a particle interpretation.

The first three layers provide a mathematically controlled precursor to spinorial physics. They do not silently supply the fourth. Later sources do close a specified free part of that fourth layer on the framed product category; Section 13 explains this without importing an interacting theory.

1.1 Object picture: why a double cover appears

Consider a continuous path of spatial frames describing one full 2π rotation. Its endpoint in $SO(3)$ is the identity, but the path is not contractible. When lifted to $\text{Spin}(3)$, that path begins at $+1$ and ends at -1 . Repeating the rotation produces a lifted path ending at $+1$. An ordinary vector representation sees only the $SO(3)$ endpoint and cannot distinguish the two loop classes; a spinorial carrier retains the sign.

This is the exact content of “loop memory” in the theorem below. It does not say that a physical object literally rotates through an extra coordinate, and it does not derive Fermi statistics or a Dirac equation. It says that the smallest connected faithful carrier for this homotopy information is the double cover.

1.2 The finite double return in one line

The q79 Fourier operator satisfies $J_{\text{FM}}^2 = -I$. The two admissible shared-line roots contribute scalar phases $+i$ or $-i$, whose squares are also -1 . Therefore

$$(\pm i J_{\text{FM}})^2 = (-1)(-I) = I.$$

This is the finite SpinC–Fourier return. The two signs are complex-conjugate lifts of the same determinant structure. The theorem does not choose one as the observed branch and does not identify the compact phase with time.

1.3 Argument map

Sections 2–4 construct the local comparison carrier and its conditional spin lift. Sections 5–8 build the selected q79 rank flag, shared line, binary sheet lift, and Clifford return. The CLN discussion then separates geometric roles from physical realizations. Sections 10–13 explain the later finite, endpoint, and free-carrier constructions. The physical-interface and numerical-program discussions distinguish their remaining hypotheses.

2 Status and dependencies

This paper is a conditional carrier theorem. Its conclusions depend on the following data, none of which follows from a bare projection map:

1. oriented Euclidean rank-three bundles TP and TI over a declared base B ;
2. a comparison field $Q_{\text{WW}} \in \Gamma(\text{Hom}(TP, TI))$;
3. an $SO(3)$ frame or holonomy bundle whose nontrivial loop class is to be retained by the state description;
4. for the q79 specialization, the selected degree-three spectral cover, its trace map, and its sheet monodromy; and
5. for a physical fermion interpretation, a separate Lorentzian spin or Spin^c completion, action, observable map, and normalization.

The terms “forced” and “necessary” below always mean necessary relative to these premises. They do not mean that MTT has selected the premises uniquely.

3 World-in-world comparison data

Definition 3.1 (Comparison field). Let $TP, TI \rightarrow B$ be oriented Euclidean vector bundles of rank three. A world-in-world comparison field is

$$Q_{\text{WW}} \in \Gamma(\text{Hom}(TP, TI)).$$

At $b \in B$, choices of oriented orthonormal frames represent $Q_{\text{WW}}(b)$ by a matrix in $\text{Mat}(3, \mathbb{R})$.

Proposition 3.2 (Dimension and component bookkeeping). *The total space of $TI \rightarrow B$ has dimension $\dim B + 3$. If $\dim B = 3$, its dimension is six. The number nine is instead the fiber dimension of $\text{Hom}(TP, TI)$.*

Proof. A rank- r vector bundle over a d -dimensional base is locally $U \times \mathbb{R}^r$ and therefore has total dimension $d + r$. Furthermore, $\text{Hom}(\mathbb{R}^3, \mathbb{R}^3)$ has dimension nine. $\square$

At an invertible, orientation-preserving background, polar decomposition gives $Q_{\text{WW}} = RU$ with $R \in SO(3)$ and U symmetric positive definite. Its component space, normalized at the identity, has the orthogonal Frobenius decomposition

$$\text{Mat}(3, \mathbb{R}) = \mathfrak{so}(3) \oplus \text{Sym}(3, \mathbb{R}), \quad 9 = 3 + 6.$$

This is not orthogonality of the two polar-coordinate variations in the ambient metric at every background: at $Q = U$, these are ΩU and δU , which can have nonzero Frobenius pairing. A metric on general polar coordinates must be pulled back explicitly; the component count survives. After a flag is selected, the symmetric part further decomposes as

$$\text{Sym}(3, \mathbb{R}) = \mathbb{R}I_3 \oplus \mathcal{D}_0 \oplus \mathcal{O}, \quad 6 = 1 + 2 + 3,$$

where $\mathcal{D}_0$ is traceless diagonal and $\mathcal{O}$ is symmetric off-diagonal. The latter split depends on the selected flag; only the $1 + 5$ trace/traceless split is invariant under the full orthogonal group.

Consequently

$$1 + 3 \times 3 = (1 + 3) + (1 + 2 + 3) = 4 + 6 = 10$$

is an exact component identity once an ordering scalar is added. It is not a proof of a ten-dimensional manifold, a $3 + 1$ spacetime, or Lorentzian signature.

4 Conditional spinorial lifting theorem

Let $F^+(TI) \rightarrow B$ be the oriented orthonormal frame bundle. Its structure group is $SO(3)$ and

$$\pi_1(SO(3)) \cong \mathbb{Z}_2, \quad 1 \longrightarrow \{\pm 1\} \longrightarrow \text{Spin}(3) \longrightarrow SO(3) \longrightarrow 1.$$

Definition 4.1 (Neutrality with loop memory). A carrier has neutrality with loop memory when its projected $SO(3)$ frame returns after a closed orientation loop while its state representation still distinguishes the two homotopy classes of such loops.

Theorem 4.2 (Conditional double-cover necessity). *Suppose a continuous state carrier over the $SO(3)$ frame data is required to distinguish the generator of $\pi_1(SO(3))$ and to compose consistently under concatenation of lifted paths. Then an ordinary vector representation of $SO(3)$ is insufficient. The minimal connected covering group on which the nontrivial class remains visible is the universal double cover $\text{Spin}(3) \cong SU(2)$.*

Proof. Every $SO(3)$ representation identifies a loop with its endpoint group element, so both homotopy classes of loops ending at the identity act as the identity. Path lifting to the universal cover assigns the two classes the endpoints $+1$ and -1 . Since $\pi_1(SO(3)) = \mathbb{Z}_2$, the universal connected cover is two-sheeted and no smaller connected cover can retain that class. The universal cover is $\text{Spin}(3) \cong SU(2)$. $\square$

Definition 4.3 (Proto-spinor). Relative to the preceding hypotheses, a proto-spinor is a section of an associated complex rank-two bundle

$$S_I = P_{\text{Spin}} \times_{SU(2)} \mathbb{C}^2,$$

where P_{Spin} is a chosen lift of the relevant oriented frame or holonomy data. The word “proto” records that no Lorentzian Dirac operator, statistics law, or particle ontology has yet been supplied.

The rank-two complex fiber is therefore not a two-component particle model by declaration. It is the smallest representation in which the lifted central sign can act nontrivially. Only after the bundle is connected to Lorentzian geometry and an action can its sections be interpreted as physical spinor fields.

The theorem does not prove that an internal world must be three-dimensional. For example, $SO(2)$ already has nontrivial winding, while higher-dimensional orientation groups also possess spin covers. Rank three is a selected realization premise supported here by the comparison carrier and the q79 degree-three data, not a universal dimensional-minimality theorem.

5 Selected q79 trace-split carrier

Let $\pi : C \rightarrow B$ be the selected degree-three spectral cover and set

$$\mathcal{A} = \pi_* \mathcal{O}_C.$$

The unit and trace define

$$p_{\text{cen}} = \frac{1}{3} \mathbf{1} \circ \text{Tr}, \quad \mathcal{A}_0 = \ker \text{Tr}.$$

Theorem 5.1 (Trace-split rank flag). *If $\mathcal{A}$ is finite locally free of rank three and the ground field has characteristic different from three, then*

$$\mathcal{A} \cong \mathcal{O} \oplus \mathcal{A}_0, \quad \text{rank}(\mathcal{O}, \mathcal{A}_0, \mathcal{A}) = (1, 2, 3).$$

Consequently the shared-line carrier

$$\mathcal{H}_{\text{CLN}} = L_{\text{shared}} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A})$$

has rank six.

Proof. The normalized trace projector is idempotent, has image the unit line, and has kernel $\mathcal{A}_0$. Rank additivity gives $3 = 1 + 2$, and the displayed direct sum has rank $1 + 2 + 3 = 6$. $\square$

This theorem selects a direct trace/trace-zero/full carrier. On a connected cover with transitive monodromy it does not produce a global ordering of three individual sheets.

The labels 1, 2, and 3 refer to ranks of nested algebraic roles: the central unit line, the trace-zero complement, and the full cover algebra. Their sum gives a six-dimensional fiber carrier. They are not three particle families, three spatial manifolds, or a count of quark colors unless a later source map proves such an identification.

6 Universal shared differential line

Let $\mathcal{L}_{64}^{\text{univ}} \rightarrow B_{\nabla} \mathbb{Z}_{64}$ be the universal flat line for the primitive character $\chi_1(n) = \exp(2\pi i n/64)$. The unique nontrivial homomorphism

$$h_{S_3} : S_3 \longrightarrow \mathbb{Z}_{64}, \quad h_{S_3}(\sigma) = 32 \epsilon(\sigma),$$

sends even permutations to zero and odd permutations to 32. Pulling $\mathcal{L}_{64}^{\text{univ}}$ back along the q79 sheet classifying map gives $(L_{\text{sh}}, \nabla_{\text{sh}})$. For both admissible odd roots $r \in \{1, 33\}$,

$$\chi_r \circ h_{S_3} = \text{sgn}.$$

Hence L_{sh} and the SpinC determinant sign line are canonically parallel-isomorphic. This equality preserves connection and holonomy; it is stronger than recognizing the same order-two phase pattern.

Theorem 6.1 (Shared-line finite naturality). *The line L_{sh} tensors the trace/trace-zero/full carrier and acts by scalar holonomy. It therefore commutes with all three lane projectors, the root-plane quarter-turn J_{DE} , and the normalized finite Hessian*

$$H_{\text{fin}} = \kappa_{\text{fin}}(I - P_{\text{Haar}}), \quad P_{\text{Haar}} = \frac{1}{6} \sum_{g \in S_3} \rho(g).$$

On the selected two-copy sheet/edge representation,

$$\text{rank } P_{\text{Haar}} = 2, \quad \text{spec}(H_{\text{fin}}/\kappa_{\text{fin}}) = \{0^{\times 2}, 1^{\times 4}\}, \quad H_{\text{TT}} = \kappa_{\text{fin}} I_2.$$

Thus the determinant, CLN, root-plane, and finite-Hessian occurrences are coherent pullbacks of one differential line, with no new fit parameter.

Proof. The determinant statement is the displayed character identity. Scalar holonomy commutes with every sheet operator. Haar averaging is an orthogonal projector, and direct decomposition of the selected representation gives the rank and spectrum. Tensoring by L_{sh} preserves all identities. $\square$

This theorem is flat and finite-symbol exact. The full physical HYM carrier has nonzero characteristic curvature and cannot literally have this flat connection. Its promotion requires a spectral-symbol functor and a parallel continuum Hessian comparison.

7 Signed sheet monodromy and its lift

For a permutation $\sigma \in S_3$, let P_σ be its permutation matrix and define

$$\rho_+(\sigma) = \text{sgn}(\sigma)P_\sigma.$$

Since $\det(\text{sgn}(\sigma)P_\sigma) = 1$, this is a representation $S_3 \rightarrow SO(3)$.

Theorem 7.1 (Local binary-dihedral lift). *The inverse image of $\rho_+(S_3)$ in $\text{Spin}(3)$ is the binary-dihedral group Dic_3 of order twelve. Lifts q_1, q_2 of adjacent transpositions may be chosen so that*

$$q_1^2 = q_2^2 = -1, \quad q_1 q_2 q_1 = q_2 q_1 q_2, \quad (q_1 q_2)^3 = -1.$$

Proof. The signed permutation image is the rotational symmetry group of an oriented triangular frame, isomorphic to the dihedral rotation group of order six. The inverse image of a finite rotation group under the two-to-one map $SU(2) \rightarrow SO(3)$ has twice its order and is its binary counterpart. Standard half-angle quaternion lifts of the two generating rotations satisfy the displayed relations, identifying the inverse image with Dic_3 . $\square$

7.1 Global obstruction

The preceding theorem is not yet a strict global Spin theorem. Let B° be the branch complement and let $\rho : \pi_1(B^\circ) \rightarrow SO(3)$ be the signed sheet monodromy. A strict lift is a homomorphism $\tilde{\rho}$ making

$$\begin{array}{ccc} & \text{Spin}(3) & \\ & \downarrow & \\ \pi_1(B^\circ) & \xrightarrow{\tilde{\rho}} & SO(3) \end{array}$$

commute. It exists exactly when the pulled-back central extension is trivial, equivalently when the relevant w_2 obstruction vanishes. Computationally, one must lift a presentation of $\pi_1(B^\circ)$ and verify that every relator closes with central sign $+1$. Local braid relations alone do not decide all global relators.

A shared $U(1)$ line can instead participate in a Spin^c cancellation of the mod-two obstruction. That is a distinct theorem and must not be reported as a strict Spin lift.

8 Clifford polarization and exact SpinC return

On the oriented two-circle exterior algebra with basis $(1, p, q, pq)$, the Poincaré-kernel transform is the integral operator

$$J_{\text{FM}} = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$J_{\text{FM}}^T J_{\text{FM}} = I, \quad \det J_{\text{FM}} = 1, \quad J_{\text{FM}}^2 = -I.$$

Thus J_{FM} is an orthogonal complex structure, equivalently one $\text{Cl}_{0,1}$ generator, and

$$P_\pm = \frac{1}{2}(I \mp iJ_{\text{FM}})$$

are complementary complex rank-two polarizations. The same derived operator reproduces the selected BHT and two-circle Hori fiber transforms.

The half-turn character of S_3 cannot itself cancel J_{FM}^2 : an involution maps to 0 or 32 in $\mathbb{Z}_{64}$, while the square roots of 32 are 16 and 48. The correct domain is the binary sheet group Dic_3 already obtained above.

Theorem 8.1 (Binary-sheet SpinC–Fourier return). *There are exactly two homomorphisms*

$$\tilde{h}_{\pm} : \text{Dic}_3 \longrightarrow \mathbb{Z}_{64}$$

which send the binary central element to 32 and the two braid generators to (16, 16) or (48, 48). Under either odd character these phases are $+i$ and $-i$, respectively. In the Fourier frame the binary transposition lift is J_{FM} , and therefore

$$U_{\pm} = \pm i J_{\text{FM}}, \quad U_{\pm}^2 = I.$$

The two lifts are complex conjugates with the same determinant sign line. No continuous or discrete fitted parameter is introduced.

Proof. The binary relations force both generator images to be equal square roots of 32; exhaustive solution in $\mathbb{Z}_{64}$ leaves only 16 and 48. The real spinor representation identifies the binary transposition with J_{FM} . Multiplication then gives $(\pm i J_{\text{FM}})^2 = I$. $\square$

This closes the double return at representation and flat differential-line tier. It does not identify compact phase with physical time, choose one of the conjugate orientations as the observed universe, or supply the nonflat physical FM/HYM correspondence.

The role of SpinC is precise here. The spin lift contributes a central minus sign and the shared $U(1)$ root contributes the compensating phase, so the combined finite operator returns exactly. A strict Spin theorem would have to make the spin part close without that $U(1)$ compensation on every global relator. This is why the finite return is a genuine result without resolving the separate w_2 obstruction.

9 Circle, lens, and nil as typed carrier roles

In this revision the three labels have the following nonexhaustive roles:

Circle. The common line L_{shared} stores phase or holonomy data. It is counted once and is not physical time.

Lens. Finite quotient, signed-sheet, or projective transport data. This does not require the global compactification to contain a literal lens-space factor.

Nil. Anchoring, upper-triangular transport, or termination data in a selected operator filtration. This does not prove a literal Nil_3 factor.

Thus circle–lens–nil is a carrier and obstruction taxonomy. The auxiliary model $L(3, 1) \times \text{Nil}_3$ may realize some roles, but it is not the selected q79 Fu–Yau compactification and the two manifolds are not identified.

Boothby–Wang geometry gives a precise supporting model for this typing. A prequantum circle bundle over $\mathbb{CP}^1$ with first Chern number k has total space $L(k, 1)$, while an integral circle bundle over T^2 has a Heisenberg nilmanifold as total space. Thus $L(3, 1)$ and Nil_3 are parallel circle-bundle realizations over different bases and curvature classes, not nested spaces [9, 10]. Their curved connections are not the flat q79 root-stack connection. All three may map to the universal classifier $BU(1)_{\nabla}$, but an MTT identification requires selected classifying maps and coherent comparison data.

10 Finite operators on the correctly typed carrier

The local six-dimensional closure carrier, qutrit response space, and finite SM Hilbert space serve different purposes. The frozen superconnection sources make their relation explicit. Frozen citations added in this edition refer to commit `f141a20e`; full immutable addresses are in the bibliography. The older managed evidence snapshot below is retained as an earlier reference, not as the authority for present frontiers.

10.1 A neutral baseline is not the gauge group

The A33 seed uses the cyclic shift $X^3 = I$ and $Y = I + X$ in the odd self-adjoint block $D = \begin{pmatrix} 0 & Y \\ Y^* & 0 \end{pmatrix}$. Its Gram is $YY^* = I + \mathbf{1}\mathbf{1}^T$: the common family line has eigenvalue 4 and its orthogonal plane has eigenvalue 1. The normalized radial curvature doubles these values. This is an exact dimensionless neutral transfer, not the profile neutrino mass matrix. Its grading-preserving unitary stabilizer is $U(1) \times U(2)$, reduced to $U(1)^3$ when the same shift must also be preserved. Neither group derives the SM gauge group from the size of a spinor block [13].

The two-stage finite branching. The stronger result uses the already selected algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}) \oplus \mathbb{C}_N$, its representation and real structure, not D alone. The two independent abelian anomaly rows select the primitive direction $(3, -1, 3)$ and charges $6Y = (1, -4, 2, -3, 6, 0)$ in all-left-Weyl order $(Q, u^c, d^c, L, e^c, N^c)$. The faithful group is $(SU(3) \times SU(2) \times U(1)_Y)/\mathbb{Z}_6$; a nonzero neutral vector in the retained Higgs doublet leaves $(SU(3) \times U(1)_{\text{em}})/\mathbb{Z}_3$, with $Q = T_3 + Y$. The A33 block injects isometrically into the three-family L_L – N_R particle incidence channel of the 96-dimensional KO6 carrier; its real structure supplies the antiparticle copy. Here N_R and N^c reflect different particle versus all-left-Weyl conventions. This compatibility does not identify the sparse injection with the complete profile-valued Dirac operator [14].

10.2 Why the common space has dimension 864

The common space is

$$F_{\text{up}} = \mathbb{C}_{\text{family}}^3 \otimes \text{HS}(\mathbb{C}_{\text{qutrit}}^3) \otimes H_{32}^{\text{SM}}.$$

The response factor has dimension 9, and the preprojection dimension is $3 \cdot 9 \cdot 32 = 864$. It is not an identification of a 27-dimensional response space with the 96-dimensional SM space. With the selected metric $\langle A, B \rangle = \text{Tr}(A^*B)/3$, the maps

$$E(\psi) = I_3 \otimes \psi, \quad T(A \otimes \psi) = \frac{1}{3} \text{Tr}(A)\psi$$

(with the family identity understood) satisfy $T = E^*$, $TE = I_{96}$ and $ET = P_{\text{center}}$. The Weyl-conjugation Laplacian has spectrum $0^1, 3^4, 6^4$, so its kernel canonically selects the scalar matrix, while its reduced Green operator retains the eight nonidentity response modes.

The selected phase and shift rows compress with coefficients $2/3$ and $1/3$, routed to $\{u, e\}$ and $\{d, \nu_D\}$ respectively. Their second-order Green Gram is $\frac{1}{27} \begin{pmatrix} 7 & 1 \\ 1 & 5 \end{pmatrix}$, with determinant $34/729$. The two response directions survive even though bare nonidentity Weyl modes have zero first-order trace. The raw upper insertion is not Hermitian; closing its orbit under adjoint and the selected KO6 real structure gives the self-adjoint odd D_{fin} with the same compression and inherited first-order algebra relations. Its rank is 768, its kernel dimension is 96, and the nonzero squared gap is the root α satisfying

$$16\alpha^3 - 32\alpha^2 + 12\alpha - 1 = 0, \quad 59/500 < \alpha < 119/1000.$$

The two occurrences of 96 must not be conflated: P_{center} is not the kernel projector of D_{fin} ; its compressed operator is full rank. The finite heat operator describes repair on this declared carrier, not a particle mass scale [15].

11 What the continuum construction actually requires

11.1 Overlap, metric, and the retained spectral window

At one selected endpoint, let the columns of $W : F_{\text{up}} \rightarrow \text{Dom}(H_{\text{cont}})$ be geometrically supplied representatives. Injectivity fixes its polar isometry $U = W(W^*W)^{-1/2}$. Equal dimensions do not select W ; the polar step normalizes the map already supplied. For a nonnegative self-adjoint dimensionless continuum Hessian on declared domains, $R_H = H_{\text{cont}}U - UH_{\text{fin}}$ controls its Galerkin compression. The sufficient bound $\|R_H\| < 59/1000$ separates a retained cluster of 96 modes from the other 768. It does not call that cluster an exact continuum kernel [16].

The residual budget includes the base operator, augmented coupling, adjoint leakage, transported metric, scale, and omitted tails. Nonnegative self-adjointness gives heat and positive-resolvent defects at most $t\|R_H\|$ and $\|R_H\|/s^2$ for $t \geq 0$, $s > 0$. Full ambient spectral coverage additionally requires a reducing range and complement exclusion, or a Feshbach–Schur comparison with a certified complement inverse. Even exact retained intertwining does not exclude unrepresented ambient zero modes. A quadratic-cost pullback alone supplies neither this domain statement nor a reducing isometry.

Products need their own compatibility. Writing $W = US$ exposes the positive distortion S . For an r -linear operation with separately typed inputs and output, the polar defect is

$$\begin{aligned} E_{U,r} &= E_{W,r}(S_1^{-1} \otimes \cdots \otimes S_r^{-1}) + U_{\text{out}}\Delta_{S,r}, \\ \Delta_{S,r} &= S_{\text{out}}m_r^{\text{fin}}(S_1^{-1} \otimes \cdots \otimes S_r^{-1}) - m_r^{\text{fin}}. \end{aligned}$$

Exact raw multiplication transport need not survive polar normalization. For connections the additional term is $-U(\nabla_{\text{fin}}S)S^{-1}$. Integrating its norm together with the transported raw connection defect controls holonomy on selected paths; nonlinear expansions also require convergence and a certified omitted tail. Linear symmetry equivariance alone is insufficient [17].

A nearby solution, when exact transport fails. The optional fixed-point result starts with the transported finite solution y_0 and complete continuum residual Φ . A bounded preconditioner B must be residual-faithful on the chosen ball: $B\Phi(y) = 0$ must imply $\Phi(y) = 0$. This matters for rectangular gauge-fixed systems. With residual bound ϵ , $\|B\| \leq M$, linear defect Z_0 and derivative Lipschitz bound L , the gates

$$M\epsilon + Z_0r + \frac{1}{2}Lr^2 \leq r, \quad Z_0 + Lr < 1$$

give one zero in that ball. A source symmetry fixing the centre and preserving the ball fixes the corrected solution by uniqueness. This is local uniqueness, not uniqueness over all endpoints; no Newton correction is needed when nonlinear transport is already exact [18].

11.2 Selecting modes and resolving multiplicities

Intrinsic Riesz projectors of the native endpoint Hessian select spectral windows before the Čech embedding. An ambient Fourier cutoff does not replace them. After source-owned commuting operators resolve multiplicities, Schur’s lemma makes each matching irreducible intertwiner one-dimensional. Polar normalization removes nonzero overlap magnitudes, but not phases. On the internal $E[3]$ blocks, product compatibility leaves nine character choices; the two selected generator anchors fix one. The rank-102 deformation bundle has fiber ranks $3 + 8 + 8 + 80 + 3$, not 864 spectral modes. A full-864 construction and a rank-96 construction with controlled Feshbach complement are different accepted routes [19].

The hidden multiplicity alternative. Qutrit symmetry on $\mathbb{C}^3 \otimes \mathbb{C}^3$ alone has commutant $M_3(\mathbb{C}) \otimes I_3$ and cannot choose the hidden multiplicity frame. For two connection-preserving unital complex star-algebra identifications, their ratio is an inner automorphism. It is trivial if the chosen lifted holonomy has scalar centralizer with the lift preserved, or if the adjoint holonomy has trivial projective centralizer. These are distinct sufficient tests. Stability or simplicity alone need not exclude discrete projective self-twists. This alternative can replace simple-spectrum assumptions, but does not manufacture the physical holonomy data [20].

12 From a selected endpoint to literal bundle data

12.1 An orbit is not yet a chosen endpoint

The endpoint atlas labels nine members of the selected free $E[3]$ graph-Prym orbit. Their physical cover class is $9H + 3D_0$, not the auxiliary $3H + 3D_0$ class. The ordinary Prym divisor and line give visible target rows $\text{rk } V_3 = 3$, $(c_1, c_2, c_3) = (0, 9u, 6\Omega)$; those rows alone are not an α -twisted line cocycle. The hidden rank-nine projective holomorphic object and existential projective HYM connection are established. What is absent is a common selected visible endpoint, explicit common chamber and literal local metrics and connections, not another hidden-existence proof.

Once local transitions, metric, connection A and normalized cover partition are supplied, the Čech isometry U gives

$$p = UU^*, \quad A_0 = U^*dU, \quad \Gamma = U(A - A_0)U^*, \quad (pd + \Gamma)U = U(d + A).$$

The Grassmann connection A_0 is not automatically the physical HYM connection. The correction records the missing difference. All inputs must descend from one immutable endpoint; rational witness arrays demonstrate the construction but cannot replace its physical inputs [21].

The full root selects the member. The later result combines member selection and the Deligne test. The 249-coefficient source has incidence rank 126, leaving 123 affine directions; one is radial rescaling and the projective tangent has dimension 122. The 248×122 response has a unit maximal minor in the selected 101-adic residue class. Under the same-source analytic Hensel hypotheses, solving its pivot rows gives a unique local candidate. All 248 coordinates of β_C must vanish on that candidate before it is the physical root. The 126 normal compatibility equations alone are not a nonlinear zero. The frozen Schur graph is not invariant under selected transport, so static linear back-substitution cannot replace the moving Gauss–Manin and connection terms. The certified very local eta9 transport is retained; it is not a useful-width continuation or an emitted physical root [22].

12.2 Class zero, trivialization, and twisted line

Vanishing of β_C establishes existence of a differential null-homotopy, not a preferred one. In the source conventions let the gerbe be $(\alpha_{ijk}, A_{ij}, B_i)$, its chosen null-homotopy (b_{ij}, a_i) , and the ordinary Prym line (ℓ_{ij}, c_i) . Then

$$\lambda_{ij} = b_{ij}\ell_{ij}, \quad q_i = c_i - a_i, \quad dq_i + B_i = dc_i$$

give the twisted line and its corrected curvature. Norm neutrality preserves the selected ordinary Prym norm and topological rows. Different null-homotopies differ by a differential line, so period-coordinate membership does not emit a cocycle without literal chain lifts [23].

Normalization within one integral branch. Collecting Deligne, norm, framing and equivariance equations as $Ch = y$, a feasible system and selected positive cochain metric M give the unique minimum-norm solution

$$h_* = M^{-1}C^*(CM^{-1}C^*)^{-1}y.$$

The inverse is on independent constraint rows; redundant rows require the corresponding Moore–Penrose inverse, and the Hilbert version requires closed range. Every other solution is $h_* + z$, $z \in \ker C$, with squared norm $\|h_*\|_M^2 + \|z\|_M^2$. This selects the continuous representative inside the supplied integral logarithm branch, not the branch or metric itself [24].

Transport also determines the target metric. For an exact invertible chain map T on the selected subspace and source metric G , the target metric is $(T^{-1})^*GT^{-1}$ and the target constraint is $C_{\text{source}}T^{-1}$. The two minimum-norm solutions obey $h_{\text{target}} = Th_{\text{source}}$. A fresh Euclidean metric in target coordinates would generally change the result. This removes a metric choice once source data are given; it does not prove global analytic invertibility of BHT [25].

The discrete branch comes from a marked path. On a selected smooth oriented interval, assume the Deligne quotient is the declared covering by its integral period local system. A starting lift and Gauss–Manin marking determine one continuous lift. If its endpoint class vanishes, the endpoint lift is an integral period in that marking. Literal period cochains then turn its integer coordinates into the normalization input above. Changing the path or starting frame can change that integer lift. The local A1 double-traversal logarithm is not a starting trivialization for the full eta9 path without a same-family comparison [26].

These constructions consume, and do not reprove, the integral ambient quotient on $V^\vee$, not an unqualified quotient on V . Local T70–T73 data do not establish global Picard/BHT descent or an ordinary Neron model. The corrected CBF T69 boundary and structural identities of the Cohesive paper, version 14, remain in force; neither a local chart nor a formal null-homotopy construction supplies the omitted global hypotheses.

13 The trace plane and the complete free carrier

13.1 A Clifford module from multiplication, not rank alone

On the selected unbranched sheet system put $W = \ker \text{Tr}$. Its determinant is the sheet-sign line because the unit line in $\mathcal{A} = \mathbb{C}\mathbf{1} \oplus W$ is fixed. Thus $\det W = L_{\text{sh}}$ with exponent one, not a fitted square-root exponent. The oriented real sheet carrier is $E_{\text{orient}} = \det(\mathcal{A}_{\mathbb{R}}) \otimes \mathcal{A}_{\mathbb{R}}$,

whose action is precisely $\text{sgn}(g)P_g$. Compressed sheet multiplication on W , together with its metric and orientation, constructs an equivariant isometry $E_{\text{orient}} \simeq \mathfrak{su}(W)$ satisfying the Clifford relation. For example the integral basis $(1, -1, 0), (1, 0, -1)$ has Gram matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$; it is not an orthonormal pair of arbitrarily selected sheets. This identifies the operations as well as the representations. The flat connection, trace projector and finite repair Hessian are compatible with the construction [27].

Ramification and the QWW symbol. The full-monodromy root stack of local orders $2, 3, 2, 1$ carries the extended trace plane and determinant line; the ordinary coarse trace-plane bundle cannot simply be continued across its rank drop. The orientation carrier is the same ρ_+ used in the q79 strain comparison. Composing with the inherited A_{QG} QWW spatial solder closes the representation-level Clifford symbol, rather than leaving a rank-only analogy. It does not make bundles on internal and physical bases literally equal. Nor does a flat root-stack connection become a nonzero-Chern HYM or curved Levi-Civita connection; a nonflat comparison or exact contorsion correction is still needed where those connections are used [28].

13.2 Different roots live on different bases

For the external transverse $U(1)$ representation, $2m = 2$ in the integer character lattice fixes $m = 1$. Restriction to the marked $\mathbb{Z}_{64}$ fiber gives χ_1 , while χ_{33} is a finite alias solving only $2r = 2 \pmod{64}$. Orientation conjugation instead gives weight -1 and residue 63. Globally the helicity-one and TT helicity-two lines over the momentum sphere have Chern numbers -2 and -4 . They cannot be identified with the internal flat shared line. The latter’s determinant comparison and the external bundle-valued helicity source map are separate questions [34].

A spacetime-only comparison is extra structure. For a declared correspondence $\mathcal{C}$ with maps p to the oriented Cauchy support Σ and q to q79, a flat comparison requires $p^*c_{\text{phys}} = q^*h_{q79}$ in $H^1(\mathcal{C}; \mathbb{Z}_{64})$. A SpinC determinant on this spin three-manifold also requires $c_1(L) \in 2H^2(\Sigma; \mathbb{Z})$. The specified lens-space countermodels separate the tests: the half-turn on $L(6, 1)$ has Chern coordinate 3 modulo 6 and fails parity, whereas coordinate 2 modulo 4 on $L(4, 1)$ passes. Adding a free cycle can leave unfixed flat characters. These are countermodels to inference from a Cauchy coframe alone, not proposed physical topologies [35].

A sufficient associated-sheet route. A literal physical S_3 cocycle, its same-source q79 comparison and a compatible tangent solder would construct W_Σ as the associated trace plane. Its determinant, parity and normalized comparison two-cell then follow, rather than being four independent selectors. Calling the physical bundle “associated” does not provide those arrays. This route is valid but the later framed construction makes it unnecessary at the complete free-carrier tier [36].

The weaker framed route. Real symmetric QWW strain has globally ordered simple eigenvalues, so its eigenvalue cover has no full S_3 permutation monodromy. A circle also has only cyclic image in S_3 . Neither observation obstructs the internal q79 sheet system. Instead a chosen QWW spin bundle and a parallel differential root $M^2 = L_{\text{sh}}$ give $W_M = S_{\text{QWW}} \otimes M$, with $\det W_M = L_{\text{sh}}$ and $\mathfrak{su}(W_M) = \mathfrak{su}(S_{\text{QWW}})$. The binary holonomy exponents 16, 48 square to 32; they are not the weight-one aliases 1, 33. Their remaining comparison inputs are further reduced by the following complete-carrier result [37].

13.3 Why the two binary roots do not split the free theory

Work on the declared external product of the framed Cauchy support and the binary q79 root-stack factor, with all pullbacks specified. The selected finite bundle is $F_{q79} = L_{\text{sh}} \otimes H_0$, $\dim H_0 = 6$. Since the restricted four-component spin representation is two copies of S_{QWW} , the same rank-24 free carrier factors as

$$S_4|_{\Sigma} \otimes F_{q79} \simeq (S_{\text{QWW}} \otimes M) \otimes (\mathbb{C}_{\text{mult}}^2 \otimes M \otimes H_0).$$

The dimensions are $2 \cdot 12 = 4 \cdot 6$, and $\mathbb{C}_{\text{mult}}^2$ is representation multiplicity, not another time coordinate. Both root factors use the same internal pullback; their parallel square recovers the old shared line and total connection. If $M_- = M_+ \otimes \varepsilon$ with $\varepsilon^2 = \mathbf{1}$, the two changes cancel on the complete product. No independent physical root/comparison selector is needed in this declared category. Internal equivariance has not become ordinary spacetime loop holonomy [38].

Which interactions inherit root cancellation. A typed tensor expression of signed root charge q changes by ε^q , hence is root-blind precisely for even q in this formal root-typing test. This includes complete-carrier expressions and the Hom and End bundles between the factors: an asymmetric Hom insertion is not automatically root-visible. For a supplied neutral action on transported domains, the parallel identification conjugates its Hessian and functional calculus. Odd unmatched terms need an additional root-charged source or compensator. Even root parity alone does not make an expression a Lorentz or gauge scalar, select an action, or supply a Majorana operator; Majorana-type contractions here describe only root typing. Positive repair is not the physical signed action, and this result does not establish J_{DE} invariance of the physical HYM endpoint [39].

Free propagation and observables. For the selected free operators on their declared domains the parallel unitary Φ obeys $D_- \Phi = \Phi D_+$. Its support preservation and uniqueness of advanced/retarded Green operators give $G_-^{\text{ret/adv}} = \Phi G_+^{\text{ret/adv}} \Phi^{-1}$. The induced one-particle unitary gives a natural graded star-isomorphism of the local CAR nets and their even observable nets. Smooth invertible transport preserves the Hadamard wavefront condition and gives a bijection of quasifree Hadamard states. GNS representations are unitarily equivalent for corresponding transported states, not arbitrary unrelated states or spin structures. The residual overall sign acts as fermion parity and fixes even observables. The two root presentations are therefore one free observable-net equivalence class at this tier; neither a preferred vacuum nor interacting renormalized QFT is selected [40].

14 Downstream physical gates

1. A Lorentzian spinor requires a globally hyperbolic base, a spin or Spin^c structure, a tetrad, a compatible connection, and a hyperbolic Dirac operator.
2. A particle interpretation requires a physical state space, statistics, observables, interactions, and an asymptotic or local excitation criterion.
3. A mass is read from a canonically normalized quadratic action or a propagator pole. A positive closure cost alone is not a mass.
4. A Higgs identification requires a selected scalar representation and couplings. A positive Hessian does not imply a unique scalar mode.

5. A discrete spectrum requires a self-adjoint operator with compact resolvent, confining boundary conditions, or another explicit spectral compactness theorem. Nil language alone does not quantize a theory.

The framed free Dirac, CAR and corresponding-state results just described already satisfy a specified part of this list. They are not reopened by the remaining same-endpoint HYM and interacting-action comparison. Weyl, Dirac, Majorana, and twistor descriptions may be built as conditional encodings once their respective chirality, pairing, real-structure, and conformal hypotheses are supplied. They are not consequences of the rank-three comparison field alone.

15 The same-source interface theorem still required

The local strain carrier is real, whereas the displayed q79 trace carrier is complex. Their rank profiles can be compared after complexifying the former, or after selecting a compatible real form of the latter. This choice does not add spacetime dimensions. A proposed *linear identification* would have to construct

$$\mathfrak{J}_{\text{WW} \rightarrow \text{q79}} : (\mathbb{R}I_3 \oplus \mathcal{D}_0 \oplus \mathcal{O}) \otimes_{\mathbb{R}} \mathbb{C} \longrightarrow L_{\text{shared}} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A})$$

on the selected base and prove:

1. compatibility with transition functions and the selected flag;
2. isometry for the declared strain and HYM inner products;
3. covariant compatibility $\mathfrak{J}\nabla^{\text{WW}} = \nabla^{\text{q79}}\mathfrak{J}$; and
4. intertwining of the Hessian, retarded, and overlap operators used by the numerical source packets.

Until these clauses are proved, the local and q79 objects are related by a dimension-matched conditional bridge, not an identity. The universal-line and finite SpinC/Fourier theorems above close important internal squares on the q79 side; they do not supply this world-in-world-to-continuum map.

The QM owner supplies a different, curved continuum language [1, 2, 3]. A finite matrix projector $p(x)$ over $C^\infty(X)$ represents a bundle, not finitely many base modes. The explicit Čech compiler transports a supplied physical connection by the pair (p, Γ) : the Grassmann connection from p alone need not be that connection. Covariant unitary changes preserve the supplied domains, curvature and Hodge calculus; a fixed raw Fourier window need not be preserved. The physical adjoint bundle is ordinary because the scalar twist cancels in endomorphisms, not because the hidden fundamental bundle is untwisted. CLN remains a local, auxiliary or filtrational carrier, not the global q79 topology or an inserted parallel flag inside an irreducible HYM bundle.

The viable spectral comparison is nonlinear. The old relative-phase-stable linear rank-six root-space route is excluded: its S_3 character $(6, 0, 0)$ differs from the two permutation copies' character $(6, 2, 0)$. The QM owner's replacement sends a Hermitian matrix with diagonal entries d_i and cyclic off-diagonal entries z_e to

$$\Phi(X) = (d_0, d_1, d_2, |z_{12}|^2, |z_{20}|^2, |z_{01}|^2).$$

This six-coordinate strain shadow forgets the triangle phase of $z_{12}z_{20}z_{01}$. The full relative-phase quotient is generically seven-dimensional, not six; its extra coordinate is neither time

nor the shared central phase. The earlier linear contract is therefore not a claim that such a root-space embedding still exists [4].

For the derivative J on the regular locus and the full reduced Green G_Q of a supplied positive source Hessian, the effective strain Hessian is $(JG_QJ^*)^{-1}$ where the reduced covariance is invertible. This is minimization over discarded directions, not an isometric embedding. A TT-only observable must first replace J by its TT projection and then perform the shorting. The later global result covers all eight edge-support strata: a missing edge has zero first derivative but a nonnegative quadratic normal jet. Its normal cost is degree one in strain, so no ordinary full six-dimensional boundary Hessian can replace it [5]. A surjective six-dimensional derivative, or the required local C_4 lift with derivative J_{DE} , forces the regular stratum. Thus regularity is no longer an independent physical input under that stronger contract. The selected endpoint, Green operator, same-source $S_3 \times C_4$ action and action normalization remain physical obligations. The quarter-turn acts on tangents; it is not a global symmetry of the strain orthant.

16 Relation to the numerical SM program

The current calculation repositories close embedded renormalized-Standard- Model equivalence at the declared one-shared-physical-primitive/profile standard, including the finite matrix, charged Yukawa profile, electroweak, threshold, and precision transport packets. Those are valid at their stated certificate tier. They do not prove that the proto-spinor carrier uniquely selects the measured profile, and they do not close the strict no-knob or unique-observed-branch upgrade. This paper therefore uses the calculations as downstream realization evidence, not as a proof of its upstream premises.

16.1 The family plane is a coordinate split, not a mass prediction

The outer class factor is the retained family factor. Its common-line projector $P_0 = \mathbf{1}\mathbf{1}^T/3$ splits the response space as $27 = 9 + 18$ and the SM carrier as $96 = 32 + 64$, compatibly with the Weyl-center reduction. It is neither the CLN flag nor color. A Hermitian family operator in these coordinates has one scalar, four plane coordinates and four real cross-block coordinates: $9 = 1 + 4 + 4$, with no parameter reduction. The block-adjugate identity makes a cross-block spectral correction quadratic, but does not assign quarks to it. If both up and down Grams commute with P_0 , their overlap has a common invariant line, hence only $1 \oplus U(2)$ mixing. The A33 4, 1, 1 degeneracy and family-blind inner Weyl action are limitations of these mechanisms, not failures of the established profile construction [29].

Static orientation and positive magnitudes. The selected static retarded source supplies $U_{10} = I$ and $U_5 = F_3$. Its Fourier transform mixes the common line with the plane, supplying a family-active orientation. But $UU^* = I$ for any unitary U , so orientation alone gives no hierarchy. The exact compiler $H_s = U_s \text{diag}(\rho_{s,0}, \rho_{s,1}, \rho_{s,2})U_s^*$ preserves three supplied positive eigenvalues; it does not derive them. The source's positive witness is only an illustration. Its F_3 is not being renamed physical CKM [30].

The later dynamic source is one orbit. For the selected Weyl convention $F_3^*XF_3 = Z$, so $I + X$ and $I + Z$ are Fourier conjugate. The common baseline involution is also preserved, making the first Hermitian response pair Fourier conjugate with spectrum $\{-4, -2, 2\}$. The pair does not commute; its positive response densities instead have the degenerate spectrum

$\{4, 1, 1\}$. This closes the finite same-source dynamic shape relation left open by the earlier static account. Routing remains phase to u, e and shift to d, ν_D , so it cannot imply a blanket “quarks second order, leptons first order” rule [31].

Complete operator vocabulary, unspecified coefficients. The actual post-response matrices A, B , not just primitive Weyl inputs, generate $M_3(\mathbb{C})$. The nine Hermitian words $I, A, B, A^2, B^2, AB + BA, i(AB - BA), ABA, BAB$ form a real basis. Inverting their trace Gram gives a dual basis and exact coefficient extraction for any Hermitian family operator. This proves expressivity, not a source rule for those coefficients. Their simple-spectrum projectors have nontrivial, CP-sensitive overlaps; the finite quartet is not the observed CKM Jarlskog invariant without the physical harmonic and sector comparison [32].

Why changing coordinates does not remove the nine rows. For $p(H) = c_0I + c_1H + c_2H^2$ on eigenvalues $(-4, -2, 2)$, the evaluation matrix with rows $(1, \lambda, \lambda^2)$ has determinant 48 (or $48a^3$ for nonzero amplitude a). Polynomial coefficients and three spectral values are equivalent coordinates. Full invariance under both responses forces a scalar and destroys hierarchy; dependence on just the two routed lanes gives six values and forces the u and e rows to agree. Sector-resolved dependence retains nine charged values unless additional source relations are derived. This is minimality within that selected finite response class, not a universal lower bound on MTT or a revision of the adopted shared-primitive/profile ledger. Positivity constrains actual positive operators, not logarithmic response coordinates [33].

16.2 What the exploratory simulation illustrates

The frozen Vue simulation exposes phase, recurrence, closure-cost and anchoring variables and lets their modeled couplings evolve. It illustrates the difference between a continuous carrier and a chosen display or composite rule. Reading its implementation also fixes its evidential limit: `smKindForSeed` assigns species, `smGaugeState` supplies charges, and `enforceSmProjector` imposes particle/color behavior. Its overlap penalty is not a CAR derivation of Pauli exclusion; declared atom content, bond lengths and the water-angle target are inputs. The energy ledger is a toy diagnostic with damping and prescribed forces, not a calibrated conservation or binding-energy theorem [41].

The “measurement” display changes internal weights and rendering through explicit rules. It is not evidence for observer-dependent dynamics, superluminal signaling or a new measurement postulate. Physical measurement remains an ordinary physical interaction described by the established operational quantum theory. Neither visual stability nor a green diagnostic label supplies a physical source value. This import is explanatory source reading; no simulation run or new numerical prediction is claimed.

17 Conclusion

The corrected proto-spinor result is narrower and stronger. Under explicit rank-three orientation and loop-memory premises, the double cover is indeed spinorial. The selected q79 carrier has an exact $1 + 2 + 3$ trace split and an exact universal flat differential line, an exact Dic_3 lift, and an exact SpinC –Fourier return on its finite carrier. The trace-plane Clifford construction and balanced binary roots further close the specified framed complete free carrier and its observable-net equivalence. The finite superconnection, family-response algebra and conditional endpoint constructions make the next hypotheses concrete. What remains is

sharply separated: strict global Spin, physical nonflat FM/HYM and same-source Hessian comparison, selected interacting action, and strict physical source-value completion. The established finite and profile results remain at their own tiers.

Computational Evidence and Reproducibility

This earlier capsule retains its historical source labels, not current obligation counts. The contextual imports in Sections 10–16 are bound to the newer immutable citations in the bibliography and their stated scopes. The selected q=79, finite-matrix, anomaly, and HYM rows provide the concrete lower structures used at the proto-spinor interface. They do not prove the supplied Lorentzian, SpinC, worldsheet, or physical-bundle hypotheses. Other Standard Model rows are corpus context, and the strict source upgrade remains open.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A11/q79_exact_audit (*derived exact*).
Executable q=79 exact-branch audit.
- A11/q79_exact_theorem (*derived exact*).
CRT q=79 theorem on the selected exact branch.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.

Corpus-state cross-checks.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.

- A01/direct_k_higgs_row (*derived exact*).
Promoted direct $K_{\text{threshold.Omega}_H.\text{lambda}}$ row.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A13/gr_tt_support (*derived exact*).
Exact-branch internal TT support certificate; physical normalization remains open.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A01/strict_pew_row (*derived exact*).
Promoted P_{EW} source row at the declared one-shared-primitive standard.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

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